

System Model Comparison

Two-stage obstacle detection pipeline — realistic timings per format and precision level. Stage 2 covers Qwen 3.5-VL at 0.8B and 2B sizes, plus the previous-generation Qwen2-VL-2B. Hardware: NVIDIA RTX 1070 (8 GB VRAM) · Values marked 'Measured' are real benchmark results; others are estimates scaled proportionally.

STAGE 1 — Eye / Filter WITH GGUF (Quantized)									
Stage (Role)	Model	Format / Precision	Params	Est. RAM Usage	Processing Time (TTFT)	Est. FPS	GGU F?	Core Capability (What does it do?)	CoT (Reason?)
Stage 1 Eye / Filter	Florence-2-base (0.23B)	GGUF Q4_K_M 4-bit quant.	~0.23B	~130 MB	~0.8–1.2 sec Measured · RTX 1070 (Optimized)	~0.9 FPS	YES	Detects obstacles, returns Bounding Box coordinates. Task-focused, no reasoning.	No. Pre-defined labels only (e.g. <OD>).
Stage 1 Eye / Filter	Florence-2-base (0.23B)	GGUF Q8_0 8-bit quant.	~0.23B	~230 MB	~1.0–1.5 sec Est. · RTX 1070	~0.75 FPS	YES	Same detection as Q4 with marginally higher accuracy. Slightly more VRAM.	No. Pre-defined labels only.
Stage 1 Eye / Filter	moondream2-2b	GGUF Q4_K_M 4-bit quant.	~1.9B	~750 MB	~1.5–2.0 sec Est. · RTX 1070	~0.5 FPS	YES	Detects/points to obstacles AND answers short visual questions about the scene (hybrid detect + caption). Broader output than Florence-2's label-only detection.	Partial. Short free-text answers, no multi-step planning.
Stage 1 Eye / Filter	moondream2-2b	GGUF Q8_0 8-bit quant.	~1.9B	~1.2 GB	~2.0–2.8 sec Est. · RTX 1070	~0.35 FPS	YES	Same hybrid detection/captioning as Q4 with better accuracy. Noticeably heavier than Florence-2 at any precision.	Partial. Slightly more coherent short answers than Q4.
STAGE 1 — Eye / Filter WITHOUT GGUF (Full Precision)									
Stage (Role)	Model	Format / Precision	Params	Est. RAM Usage	Processing Time (TTFT)	Est. FPS	GGU F?	Core Capability (What does it do?)	CoT (Reason?)
Stage 1 Eye / Filter	Florence-2-base (0.23B)	Non-GGUF FP32 Full precision	~0.23B	~900 MB	~2.5–4.0 sec Est. · RTX 1070	~0.3 FPS	NO	Same capability — full precision weights. Much heavier on VRAM, slower on edge hardware.	No. Same label-only behavior, higher resource cost.
Stage 1 Eye / Filter	moondream2-2b	Non-GGUF FP16 Full precision	~1.9B	~3.5 GB	~4.5–6.0 sec Est. · RTX 1070	~0.18 FPS	NO	Same hybrid detect+caption capability at full precision. Best accuracy of the moondream2 variants, but far too slow for real-time obstacle avoidance on this GPU.	Partial. Full-precision short answers; latency makes it unsuitable for real-time use.
STAGE 2 — Brain / Router WITH GGUF (Quantized)									
Stage (Role)	Model	Format / Precision	Params	Est. RAM Usage	Processing Time (TTFT)	Est. FPS	GGU F?	Core Capability (What does it do?)	CoT (Reason?)
Stage 2 Brain / Router	Qwen 3.5-VL-0.8B	GGUF Q4_K_M 4-bit quant.	~0.8B	~700–900 MB	~3.0 sec Est. · RTX 1070	~0.33 FPS	YES	Analyzes Florence's coordinates + image. Plans escape route with step-by-step reasoning.	Yes. CoT: "Turn right to avoid obstacle at X,Y."
Stage 2 Brain / Router	Qwen 3.5-VL-0.8B	GGUF Q8_0 8-bit quant.	~0.8B	~1.1–1.3 GB	~4.5–5.5 sec Est. · RTX 1070	~0.2 FPS	YES	Same routing logic, better output quality than Q4. Approaching VRAM limit on RTX 1070.	Yes. Full CoT, slightly better accuracy than Q4.
Stage 2 Brain / Router	Qwen 3.5-VL-2B	GGUF Q4_K_M 4-bit quant.	~2B	~1.6–1.9 GB	~2.5–3.75 sec Est. · RTX 1070	~0.3 FPS	YES	Larger Qwen 3.5-VL variant. Same routing/planning role as the 0.8B model but with noticeably better reasoning quality at similar TTFT.	Yes. Full CoT, higher-quality plans than the 0.8B model.
Stage 2 Brain / Router	Qwen 3.5-VL-2B	GGUF Q8_0 8-bit quant.	~2B	~2.5–2.8 GB	~3.75–5 sec Est. · RTX 1070	~0.22 FPS	YES	Best output quality of the Qwen 3.5-VL sizes tested. Uses over a third of the RTX 1070's 8GB VRAM alongside Stage 1.	Yes. Full CoT, best accuracy among GGUF options.
Stage 2 Brain / Router	Qwen2-VL-2B	GGUF Q4_K_M 4-bit quant.	~2B	~1.6–1.9 GB	~2.5–3.75 sec Est. · RTX 1070	~0.3 FPS	YES	Previous-generation 2B Qwen VLM. Comparable routing/reasoning capability to Qwen 3.5-VL-2B at this quant level, included as a fallback/compatibility option.	Yes. Step-by-step CoT, similar quality to Qwen 3.5-VL-2B.
Stage 2 Brain / Router	Qwen2-VL-2B	GGUF Q8_0 8-bit quant.	~2B	~2.5–2.8 GB	~3.75–5 sec Est. · RTX 1070	~0.22 FPS	YES	Same as Q4 with marginally higher accuracy. Slightly behind Qwen 3.5-VL-2B in output quality generation-over-generation.	Yes. Full CoT, slightly weaker than the 3.5 generation.
Stage 2 Brain / Router (alt.)	moondream2-2b	GGUF Q4_K_M 4-bit quant.	~1.9B	~750 MB	~1.5–2.0 sec Est. · RTX 1070	~0.5 FPS	YES	Lightweight alternative router: can answer short questions about Florence's detections (e.g. "which side is clear?") but lacks Qwen's structured multi-step planning.	Partial. Short free-text answers, not true step-by-step CoT.
Stage 2 Brain / Router (alt.)	moondream2-2b	GGUF Q8_0 8-bit quant.	~1.9B	~1.2 GB	~2.0–2.8 sec Est. · RTX 1070	~0.35 FPS	YES	Same limited-reasoning router role as Q4 with better answer quality. Fastest router option, but weakest reasoning depth of the Stage 2 candidates.	Partial. Slightly more coherent short answers than Q4.
STAGE 2 — Brain / Router WITHOUT GGUF (FP16)									
Stage (Role)	Model	Format / Precision	Params	Est. RAM Usage	Processing Time (TTFT)	Est. FPS	GGU F?	Core Capability (What does it do?)	CoT (Reason?)
Stage 2 Brain / Router	Qwen 3.5-VL-0.8B	Non-GGUF FP16 Custom Optimized	~0.8B	~2.0–2.5 GB	~3.0 sec Measured · RTX 1070 Self-optimized (flash-attn, bf16 cast, custom loader)	~0.33 FPS	NO	Full precision — same speed as GGUF Q4 thanks to custom optimization. Superior output quality vs any GGUF variant. No quantization loss.	Yes. Full CoT at full precision. Best accuracy in the pipeline.
Stage 2 Brain / Router	Qwen 3.5-VL-0.8B	Non-GGUF FP16 Default (No Optimization)	~0.8B	~2.0–2.5 GB	~10–14 sec Measured · RTX 1070 (Non-opt.)	3–4x slower / ~0.08 FPS	NO	Default FP16 load — no flash-attn, no bf16 casting, no custom loader. Same weights as optimized version but 3–4x slower. Unusable for real-time.	Yes. Full CoT — but latency makes it completely unsuitable for real-time use.
Stage 2 Brain / Router	Qwen 3.5-VL-2B	Non-GGUF FP16 Default (No Optimization)	~2B	~4.5–5.5 GB	~6–8 sec Est. · RTX 1070 (Non-opt.)	~0.15 FPS	NO	Full precision, no quantization loss, best reasoning quality of the 2B tier. RAM usage plus Stage 1 leaves little VRAM headroom on an 8GB card.	Yes. Full CoT at full precision, but too slow for real-time without custom optimization.
Stage 2 Brain / Router	Qwen2-VL-2B	Non-GGUF FP16 Default (No Optimization)	~2B	~4.5–5.5 GB	~6–8 sec Est. · RTX 1070 (Non-opt.)	~0.15 FPS	NO	Same VRAM/latency profile as Qwen 3.5-VL-2B FP16; slightly older architecture generation. Same real-time limitations apply.	Yes. Full CoT at full precision, same latency issue as Qwen 3.5-VL-2B.
Stage 2 Brain / Router (alt.)	moondream2-2b	Non-GGUF FP16 Full precision	~1.9B	~3.5 GB	~4.5–6.0 sec Est. · RTX 1070	~0.18 FPS	NO	Full-precision router alternative. Best answer quality of the moondream2 variants, but still limited to short Q&A rather than structured multi-step route planning.	Partial. Full-precision short answers; not true step-by-step CoT.

GGUF Q4_K_M: 4-bit quantization — smallest footprint, fastest inference, minimal accuracy loss. Best for real-time edge use. GGUF Q8_0: 8-bit quantization — better accuracy than Q4, ~1.5–2× more RAM, moderately slower. Non-GGUF FP16 Optimized: full precision with flash-attention / memory optimizations enabled (measured only for Qwen 3.5-VL-0.8B). Non-GGUF FP16 Default: default FP16 load — no flash-attn, significant latency penalty. TTFT = Time To First Token. moondream2-2b figures are estimates scaled from its known lightweight architecture and public ~1.2GB 4-bit RAM footprint, not measured on this hardware; included as an alternative Stage 1 candidate since it combines detection with lightweight visual Q&A in one model. Qwen 3.5-VL-2B and Qwen2-VL-2B rows are estimates scaled from the VLM Comparison benchmark table, not measured on this hardware; included alongside Qwen 3.5-VL-0.8B to show the size/quality trade-off available at Stage 2 and to compare against the previous-generation Qwen2-VL-2B. moondream2-2b also appears as an alternative Stage 2 router: it can answer short questions about Florence's detections but is not built for structured multi-step route planning like the Qwen models, so its CoT is marked "Partial".